Anarv Vasavada

AI Systems Engineer

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Portfolio | GitHub | LinkedIn | Blog

Education

- **Parul University** — Vadodara, Gujarat, India
  Bachelor of Technology - Computer Science and Engineering; GPA: 7.75/10 — *June 2022 - Expected June 2026*
  Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Object-Oriented Programming (Java)

Skills

- **Languages**: Python, Java, JavaScript, TypeScript, C, SQL
- **AI / Machine Learning**: PyTorch, TensorFlow, YOLO, NLP, LLMs, LangChain, RAG
- **Backend & Systems**: FastAPI, Node.js, REST APIs, Distributed Systems, Redis
- **Databases**: PostgreSQL, MongoDB, SQLite, Vector Databases
- **Web & Tools**: React.js, Docker, Git, Linux, CI/CD
- **Soft Skills**: Ownership, Technical Leadership, Systems Thinking, Execution Under Pressure

Experience

- **Upsquare Technologies** — India
  Artificial Intelligence Intern — *Jan 2026 - Present*
  - **Computer Vision Pipelines**: Developed and debugged video analytics pipelines for real-world CCTV and transportation environments.
  - **Model Training and Evaluation**: Prepared datasets and conducted YOLO-based fine-tuning with evaluation across occlusion, lighting, and camera-angle variations.
  - **System Integration**: Connected AI inference workflows with backend services and a web-based upload-to-results pipeline.

Projects

- **Inflion: AI Observability Framework** — AI Infrastructure Architect
  Open Source Python Package / AI Systems Observability Project — *2025 – Present*
  - **Observability and Traceability Layer**: Designed an AI observability framework composed of 3 core components (execution tracing, behavioral analysis, and influence tracking) to monitor AI system interactions across multi-step workflows.
  - **Execution Monitoring**: Defined structured tracing pipelines applied across 100% of tracked execution flows, enabling deterministic attribution and traceability across chained agent operations.
- **Octopus: Autonomous Multi-Agent Orchestration System** — AI Systems Architect
  AI Systems Project — *2025*
  - **System Architecture**: Designed an orchestration layer coordinating workflows across 5 integrated production tools and 2 execution phases (planning and execution), replacing manual handoffs present in baseline pipelines.
  - **Memory and Reasoning**: Introduced shared vector memory and structured reasoning pipelines supporting 10+ concurrent agent executions across multi-step tasks without context resets.
  - **Automation Impact**: Observed a 35–40% decrease in end-to-end task completion time and a 2x increase in reusable workflows during repeated task executions using standardized agent templates.
- **Memron.ai: Persistent Multi-Agent Memory System** — AI Systems Engineer
  Distributed AI Systems / Blockchain Project — *2025*
  - **Persistent Agent Memory**: Engineered a distributed memory layer spanning 2 storage systems (Avalanche for verification and IPFS for persistence) to retain AI agent state across multi-step execution sessions.
  - **Cross-Agent Synchronization**: Developed synchronization protocols enabling state propagation between 5–8 concurrently executing agents with sub-second consistency delays under controlled orchestration environments.
  - **System Performance**: Improved context retrieval latency by 45–50% and reduced overwritten or lost memory events by 35–40% relative to non-deterministic baseline reads.
- **VisionX: AI-Driven Developer Collaboration Platform** — Full Stack Engineer
  Full Stack Systems Project — *2023 – 2024*
  - **Platform Development**: Built a GitHub-integrated collaboration platform consisting of 6 major modules (auth, PR validation, analytics, incentives, workflows, dashboards) used across multiple active repositories.
  - **Impact**: Recorded a 50–60% increase in contributor participation and a 40–45% reduction in manual PR review effort during pilot usage across repeated contribution cycles.

Publications

- **The Cost of Coordination: Scheduling and Consistency in Multi-Agent AI Systems (2025)**: Lead author. Designed and validated the Token-Latency-Cost Optimization (TLCO) scheduling framework alongside the HCP-A causal consistency protocol to mitigate coordination overhead and stale state in multi-agent systems.
  Results: Achieved 45% improvement in scheduling efficiency and 40% reduction in stale reads under high-concurrency simulations.